LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Practice Pack: Force and Laws of Motion

Chapter / Topic: Topic-based: use with whichever Class 9 book your students follow

For teacher use only. Do not distribute to students.

LEVEL 1: CONCEPTS

Q1. State Newton's first law of motion.

Answer: An object stays at rest, or in uniform motion in a straight line, unless an unbalanced external force acts on it.

Q2. Why do passengers lurch forward when a moving bus stops suddenly?

Answer: Because of inertia of motion. The bus stops, but the passengers' bodies tend to keep moving forward.

Q3. Define momentum and give its SI unit.

Answer: Momentum is the product of mass and velocity ($p = mv$). Its SI unit is kg m/s.

Q4. A rubber ball and a stone are the same size. Which has more inertia, and why?

Answer: The stone, because it has more mass. Inertia depends on mass.

Q5. State Newton's third law of motion with one example.

Answer: To every action there is an equal and opposite reaction, and the two act on different bodies. Example: a gun recoils backwards when a bullet is fired forwards.

LEVEL 2: NUMERICALS

Q6. A force of 5 N gives mass m_1 an acceleration of 10 m/s^2 and mass m_2 an acceleration of 20 m/s^2. What acceleration would it give if both masses were…

Answer: $m_1 = 5/10 = 0.5$ kg, $m_2 = 5/20 = 0.25$ kg. Total = 0.75 kg. $a = 5/0.75 \approx 6.67$ m/s^2.

Q7. A car of mass 1200 kg moving at 90 km/h is slowed down to 18 km/h in 4 s. Find the acceleration, the change in momentum, and the force applied.

Answer: u = 25 m/s, v = 5 m/s. $a = (5 – 25)/4 = –5$ m/s^2. Change in momentum = $1200 \times (5 – 25) = –24{,}000$ kg m/s. Force = $ma = 1200 \times 5 = 6000$ N (opposite to the motion).

Q8. An object of mass 100 kg is accelerated uniformly from 5 m/s to 8 m/s in 6 s. Find its initial and final momentum and the force applied.

Answer: Initial $p = 100 \times 5 = 500$ kg m/s. Final $p = 100 \times 8 = 800$ kg m/s. $F = (800 – 500)/6 = 50$ N.

Q9. A bullet of mass 20 g is fired from a pistol of mass 2 kg with a velocity of 150 m/s. Find the recoil velocity of the pistol.

Answer: Total momentum before = 0. So $0.02 \times 150 + 2 \times v = 0$, which gives $v = –1.5$ m/s. The pistol recoils at 1.5 m/s, opposite to the bullet.

Q10. A hockey ball of mass 200 g moving at 10 m/s is hit so that it goes back along its path at 5 m/s. Find the change in its momentum.

Answer: Take the initial direction as +. Change in momentum = $m(v – u) = 0.2 \times (–5 – 10) = –3$ kg m/s. The magnitude is 3 kg m/s, opposite to the original direction.

LEVEL 3: CHALLENGE (FOUNDATION)

Q11. Two objects of masses 100 g and 200 g move along the same line and direction at 2 m/s and 1 m/s respectively. They collide, and after the collision…

Answer: Momentum before = $0.1 \times 2 + 0.2 \times 1 = 0.4$ kg m/s. After: $0.1 \times 1.67 + 0.2 \times v = 0.4$, so $0.2v = 0.233$ and $v \approx 1.165$ m/s (same direction).

Q12. Why does a cricketer pull the hands backwards while catching a fast ball? Use the idea of force and time.

Answer: The ball's momentum has to be reduced to zero. Moving the hands back increases the time taken to stop the ball. Since F = (change in momentum)/time, a longer time means a smaller force on the hands, so they hurt less.

Q13. A truck starts from rest and rolls down a slope with constant acceleration, covering 400 m in 20 s. Find its acceleration and the force acting on it…

Answer: $s = \frac{1}{2}at^2$, so $400 = \frac{1}{2} \times a \times 400$ and $a = 2\ m/s^2$. $F = ma = 7000\ kg \times 2 = 14{,}000\ N$.